

CAPSTONE PROJECT

Detailed Project Report

A Python application that prepares a bank-ready term-loan project report, demonstrated on a 125-room hotel case study

CA Asis Kumar Das

The Problem

What the bank requires

A Detailed Project Report: project cost, means of finance, income statement, balance sheet, cash flow, and ratios such as DSCR and IRR.

Why spreadsheets fall short

One changed assumption must flow through dozens of cells. A balance sheet that does not tally can go unnoticed, and formatting takes hours.

What this project does

The user enters assumptions once. The application produces a consistent, verified report.

Objectives

01

Calculation engine

Construction phasing, interest during construction, depreciation, ten-year forecast and loan schedule.

02

Core statements

Income Statement, Balance Sheet and Cash Flow Statement.

03

Lender indicators

DSCR, IRR, payback period and break-even point.

04

Professional reports

Export to Excel and PDF.

Case study: a 125-room premium business hotel, validated against a reference financial model.

Technology Stack

Python

Core language for the engine and the interface.

Tkinter

Desktop interface. Ships with Python.

openpyxl

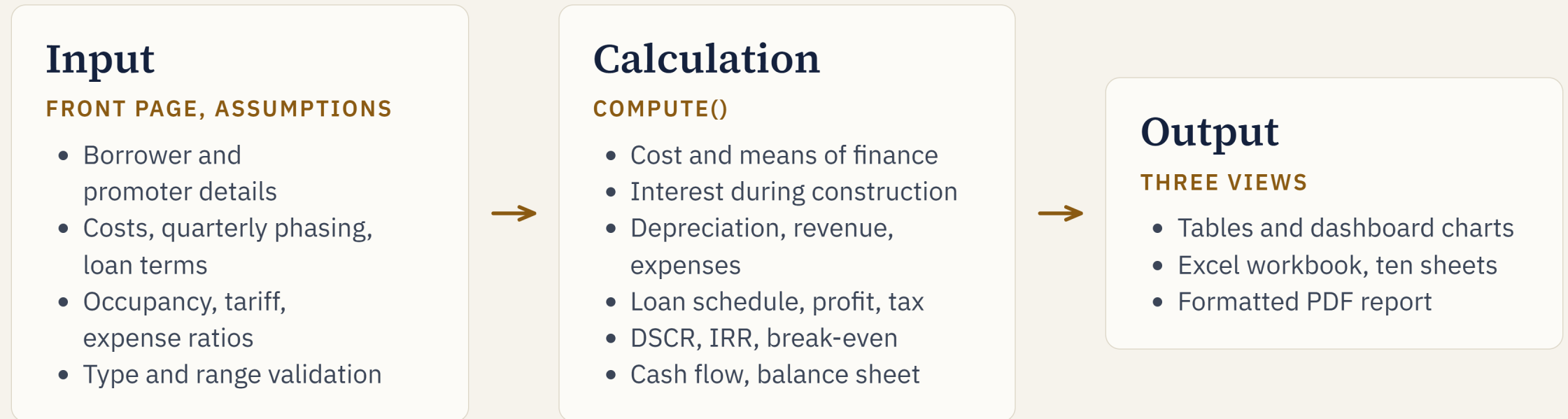
Excel export: a workbook of ten sheets.

reportlab

PDF export: a formatted bank report.

Calculations use the standard library only, including a custom bisection routine for IRR. A launcher batch file installs the two libraries and starts the program.

Three Layers, One Engine



One results dictionary feeds every view, so screen, Excel and PDF always show the same figures.

Implementation Highlights

4,800

lines of Python in the application, plus a separate test suite

Dynamic model

1 to 20 construction quarters, 1 to 25 years of tenure, cost items added or removed.

Interest during construction

Quarterly, on the average loan balance. Funded by the promoters.

Repayment profiles

Stepped (2.5% rising to 12.5% a year) or equal installments.

Verified by tests

Reference-model comparison, balance sheet, cash flow, loan and NPV-at-IRR checks.

Live Demonstration

1

Front Page

2

Assumptions

3

Generate Report

4

What-if analysis

5

Excel and
PDF export

What-if: opening occupancy lowered from 65% to 60%, then the report is regenerated.

Results: Base Case

134.58

Total project cost,
Rs. crore

70.50

Term loan, Rs. crore

1.10x

Debt to equity ratio

71.4%

Break-even in year 1,
share of capacity

1.63x

Average DSCR

1.40x

Minimum DSCR,
benchmark 1.25x

10.31%

Project IRR, loan
rate 10%

7.38

Simple payback, years

The balance sheet tallies in every year, with zero difference.

Limitations and Future Scope

Limitations

- One property type; some labels are hotel-specific
- Single base case; no scenario or sensitivity module
- Working-capital facility recorded as text, not modelled
- Windows-oriented desktop interface

Future scope

- Automated sensitivity and scenario analysis
- Working-capital assessment
- Templates for other industries
- Web version for shared use

Conclusion

One tested Python application replaces a fragile spreadsheet process: assumptions go in, and verified statements, lender ratios and reports come out.

Thank you. Questions are welcome.

CA Asis Kumar Das